

Lab 3

s353961 Xu Minghong

Part 1

We first set up the model of the patch in CST:

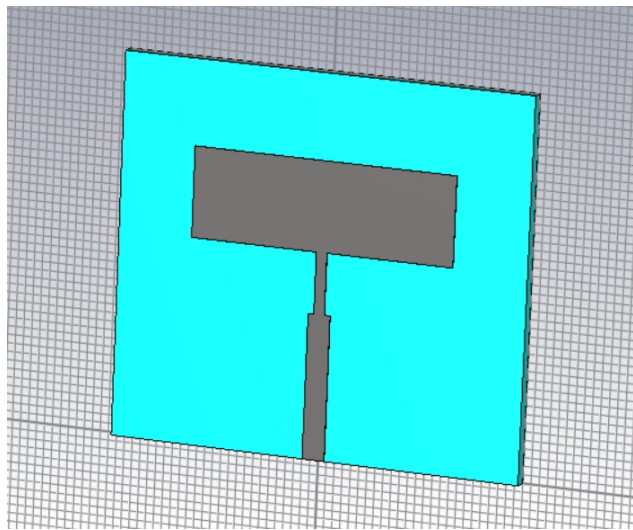


Fig.1 Layout of the antenna

then we run the solver and use Matlab to plot the figure:

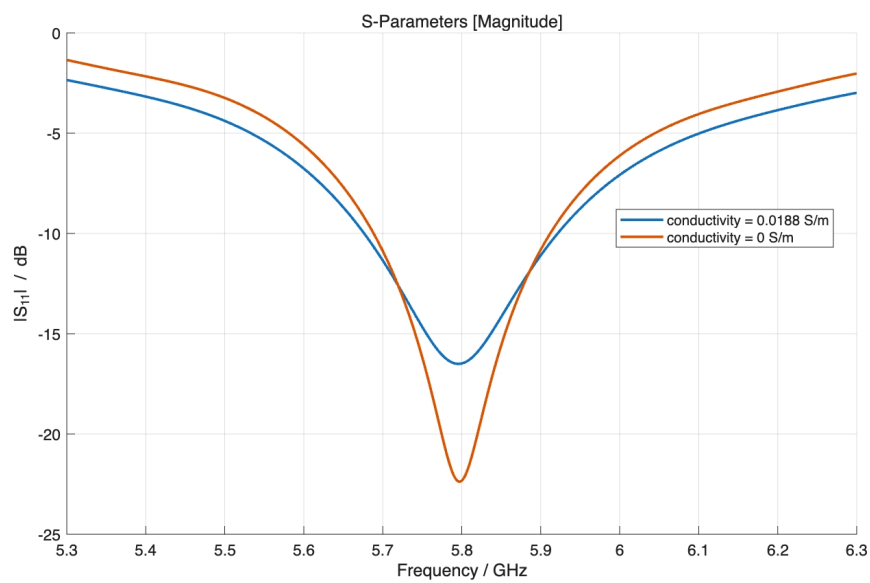


Fig.2 Behavior of $|S_{11}|$

For both cases, the minimum value appears around 5.8 GHz, which means the antenna is correctly designed at the target frequency. When conductivity = 0 S/m, the minimum $|S_{11}|$ is lower, indicating better impedance matching. When conductivity = 0.0188 S/m, the curve becomes smoother and the minimum value increases because dielectric losses reduce the antenna efficiency.

So the substrate loss mainly affects the matching quality rather than the operating frequency.

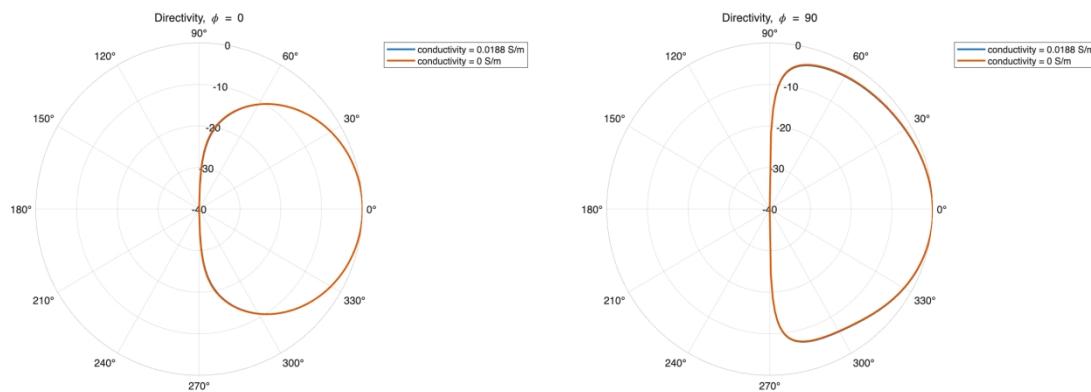


Fig.3&4 Radiation pattern

The radiation patterns for $\phi = 0^\circ$ and $\phi = 90^\circ$ show the typical broadside behavior of a patch antenna. In both planes, the maximum radiation is around 0° , which is normal because the antenna mainly radiates perpendicular to the patch surface.

From the CST radiation patterns, the two principal planes are identified as follows:

The E-plane is $\phi = 90^\circ$, because this plane contains the main radiation direction and the electric-field direction; The H-plane is $\phi = 0^\circ$, because this plane contains the main radiation direction and the magnetic-field direction.

Therefore, in this CST model:

E-plane: $\phi = 90^\circ$

H-plane: $\phi = 0^\circ$

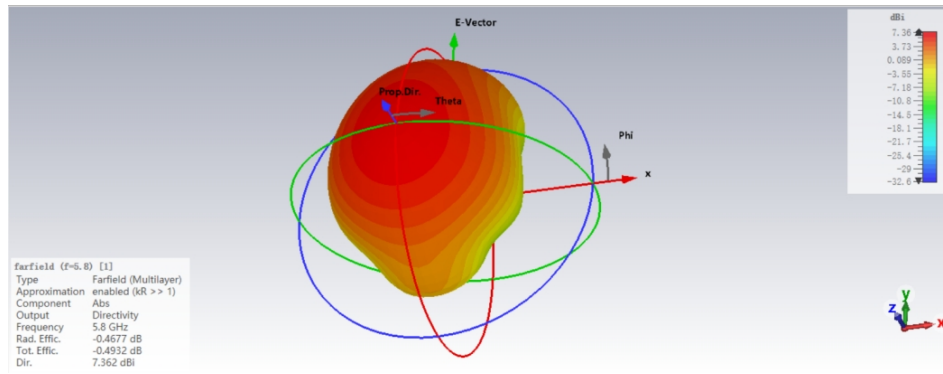


Fig.5 Polarization

The CST farfield result shows that the antenna is linearly polarized. The electric field mainly points along one fixed direction, without circular or elliptical rotation.

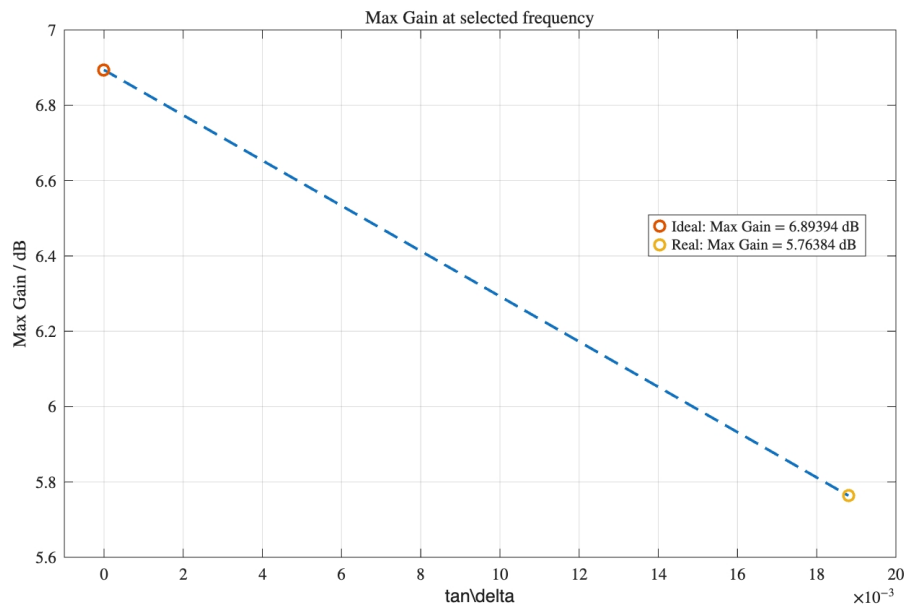


Fig.6 Max gain

The figure shows the maximum gain of the antenna for the ideal substrate and the real substrate. When conductivity = 0 S/m, the maximum gain is about 6.89 dB. When conductivity = 0.0188 S/m, the maximum gain decreases to about 5.76 dB.

This reduction is caused by dielectric losses in the substrate. Part of the electromagnetic power is dissipated inside the material instead of being radiated, so the antenna efficiency becomes lower.

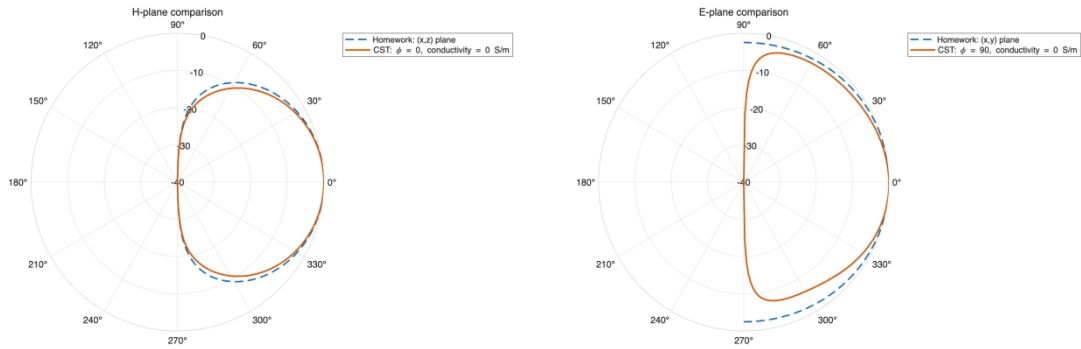


Fig.7&8 Comparison of radiation pattern

The CST radiation patterns are compared with the theoretical patterns obtained in **Assignment 3**. Because the radiation patterns for conductivity = 0 S/m and conductivity = 0.0188 S/m almost overlap, only the conductivity = 0 S/m case is used for the comparison with the analytical results.

In both the E-plane and H-plane, the main radiation direction is the same.

The CST and theoretical curves are generally consistent, especially near the main lobe region. Small differences appear in the beamwidth and side regions because the theoretical model is based on simplified assumptions, while the CST simulation includes the real antenna geometry and substrate effects.

Part 2

We use the VNA to measure $|S_{11}|$ of the antennas after calibration and export the data to plot the figure with Matlab.

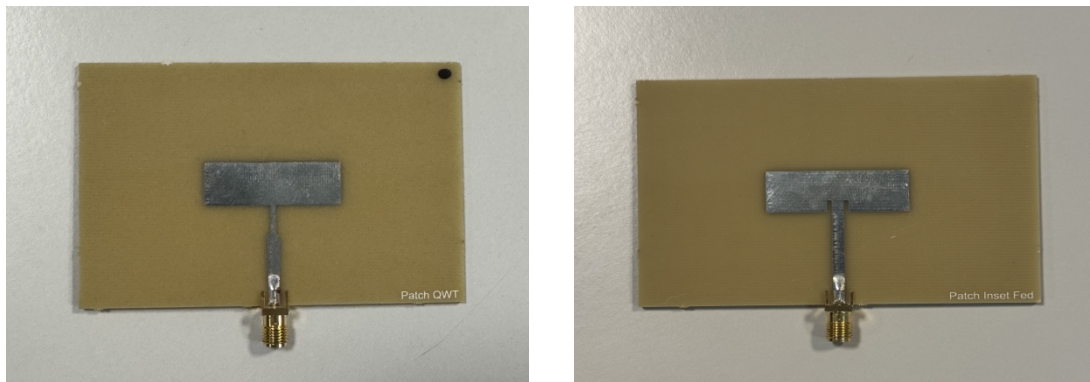


Fig.9&10 Antenna prototypes

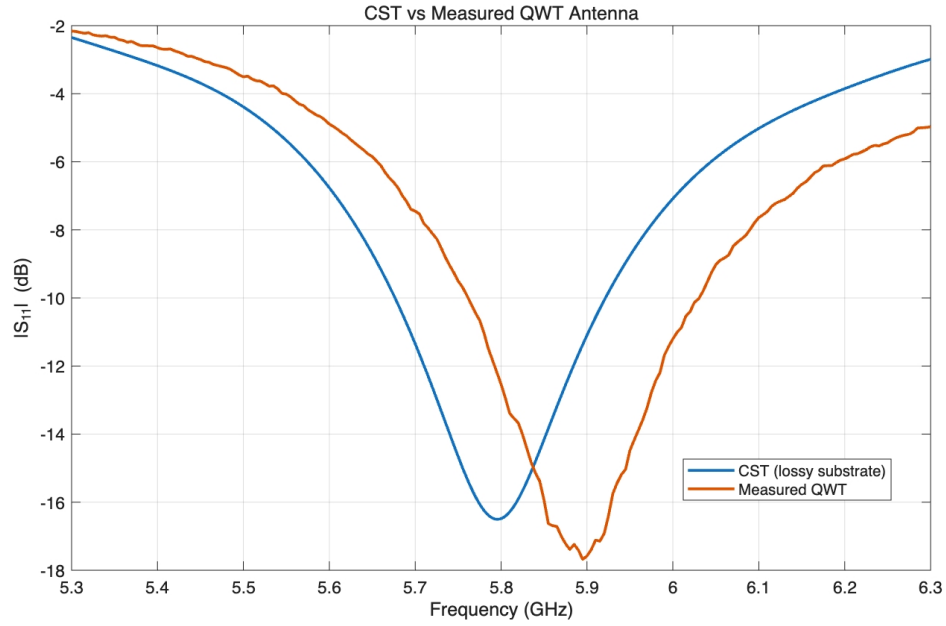


Fig.11 $|S_{11}|$ comparison between CST and measured antenna with QWT

The measured resonance frequency is slightly shifted toward a higher frequency compared with the CST result. This difference may be caused by fabrication tolerances, connector effects, substrate parameter variations and measurement uncertainties. The measured curve is also less smooth and slightly asymmetric because real measurements include losses, noise and imperfections that are not fully considered in the CST model.

However, the overall trend of the two curves is consistent, showing that the CST model gives a reasonable prediction of the antenna behavior.

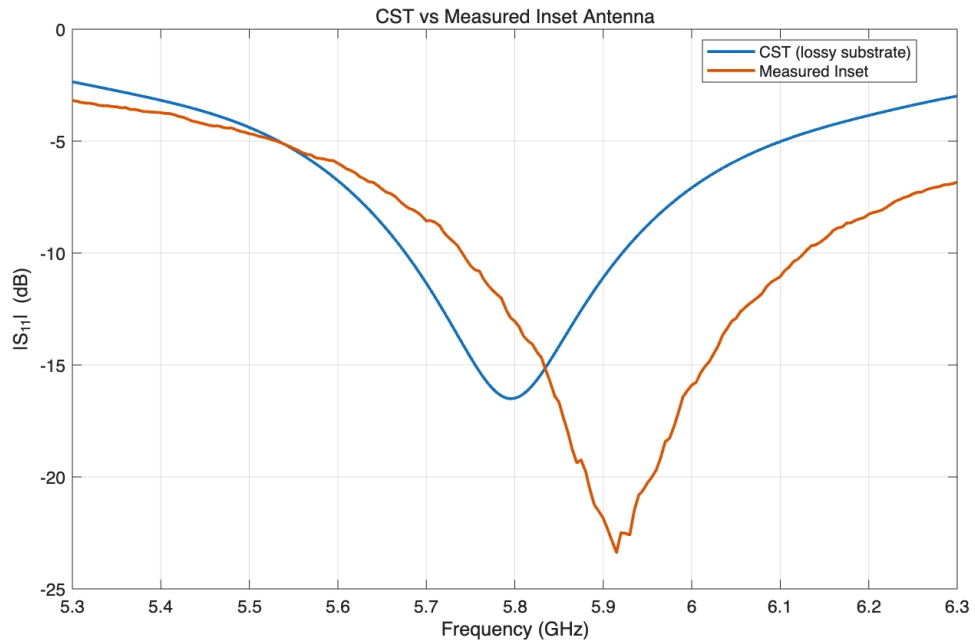


Fig.12 $|S_{11}|$ comparison between CST and measured antenna with insets

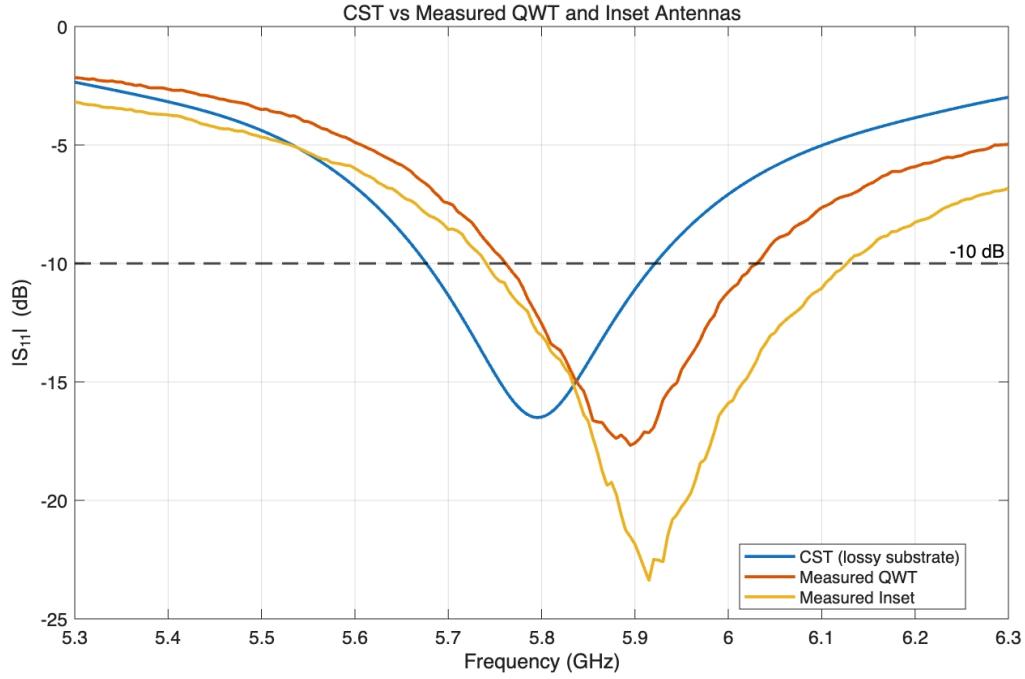


Fig. 13 $|S_{11}|$ comparison between CST and two measured antennas

Compared with the CST result, the measured resonance frequency is shifted to a slightly higher frequency and the minimum $|S_{11}|$ value is lower. This indicates that the measured antenna has stronger matching at resonance.

The differences between simulation and measurement may come from fabrication tolerances, soldering and connector effects, substrate parameter variations and measurement uncertainties.

Although some differences exist, the measured and simulated curves have a similar overall trend, showing that the CST model can predict the antenna performance reasonably well.